

Math 350 Section 1

Fall 2024

Exam II

October 23, 2024

Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) a monotone increasing sequence must converge

(b) the union of any family of open sets is open

(c) if a set is not closed, then it must be open

(d) in a metric space with the discrete metric, every set is both open and closed

(e) every convergent sequence is also Cauchy

2. (10 points)

(a) Write down the definition of a metric space.

(b) Write down three different examples of metrics we can define on $\mathbb{R}^2$.

(c) Explain why $d(x, y) = |x + y|$ is NOT a metric on $\mathbb{R}$.

3. (10 points)

(a) Write down the definition of an open set in a metric space (M, d) .

(b) Let (M, d) be a metric space and let $x \in M$ and $r > 0$. Prove that the open ball

$$B_M(x; r) = \{y \in M : d(x, y) < r\}$$

is an open set without using any theorems from class.

4. (10 points)

- (a) Write down the definition of a sequence $\{x_n\}$ of values in a metric space (M, d) converging to a value $L \in M$.
- (b) Prove that if a sequence $\{x_n\}$ converges to a value $L \in M$, then there exists a constant $R > 0$ such that $d(x_n, L) < R$ for all $n \in \mathbb{Z}_+$.

5. (10 points) Carefully state each of the following theorems.

(a) the Lindelöf Covering Theorem

(b) the Bolzano-Weierstrass Theorem

(c) the Cantor Intersection Theorem

(d) the Heine-Borel Theorem

6. (10 points)

(a) Write down the definition of an open cover of a subset $A \subseteq M$ of a metric space M .

(b) Prove that the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = (1/i, 2)$ is an open cover of the interval $(0, 1]$

7. (10 points)

(a) Write down the definition of a subset $A \subseteq M$ of a metric space M being compact.

(b) Prove that if $A = \{a\}$ for some $a \in M$, then A is compact.